

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (ME)

MAY 2018

ME308 /ME308.01 Industrial Management

Date: 10/05/2018, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Due credits will be given to own interpretations

SECTION – I

Q - 1 **Select the appropriate answers.**

[07]

- (a) Scientific management is based on
- | | |
|---|---|
| i. The scientific observation of people at work would reveal the one best way to do the task. | ii Workers can decide their own methods of performing tasks. |
| iii. Observation would reveal the workers need to be multi-skilled. | iv. Workers would receive a set wage regardless of performance. |
- (b) What is the guiding principle of Scientific management?
- | | |
|-----------------------------|---------------------------------|
| i. One best way to do a job | ii. Fluid working relationships |
| iii. Experimentation | iv. Freedom of association |
- (c) Name the founding father of scientific management theory.
- | | |
|---------------------|--------------------|
| i. Frederick Taylor | ii Frank Gilbreth. |
|---------------------|--------------------|
- (d) Work study is concerned with
- | | |
|---|---|
| i. Improving present method and finding standard time | ii. Improving production planning and control |
| iii. Motivation of workers. | iv. Improving production capability. |
- (e) What does symbol 'O' imply in work study?
- | | |
|--------------|-------------------|
| i. Operation | ii. Inspection |
| iii. Delay. | iv. None of these |
- (f) Material handling in automobile industry is done by
- | | | |
|--------------------|-------------|-------------------|
| i. Overhead crane, | ii. Trolley | iii Belt conveyor |
|--------------------|-------------|-------------------|
- (g) Break-even analysis can be used for
- | | |
|---------------------------------|---------------------------------------|
| i. Short run analysis | ii. Long run analysis. |
| iii. There is no such criterion | iv. Average of above two run analysis |

- Q - 2 (a)** Discuss use of work study in manufacturing of automobiles. [07]
- (b)** Prepare organizational chart for foundry industry and enlist role of each position in one line. [07]

OR

- Q - 2 (a)** Identify SAP softwares role in HRA. [07]
- (b)** As a trainee engineer how do you behave in a company for seeking training? [07]
- Q - 3 (a)** Identify layout and elaborate site selection factor for same as in figure 01. [07]

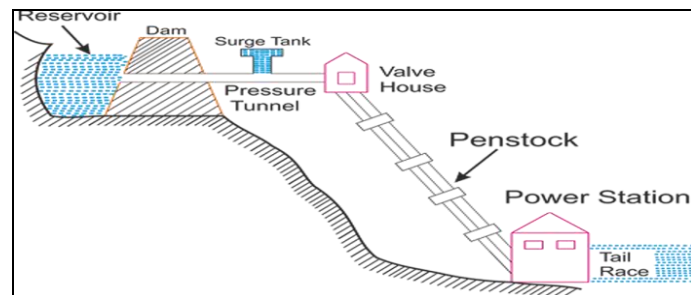


Figure 01 Layout

- Q - 3 (b)** Elaborate use of conveyors, cranes, and industrial trucks for transport with respect to their path and area of operation as in figure 02. [07]

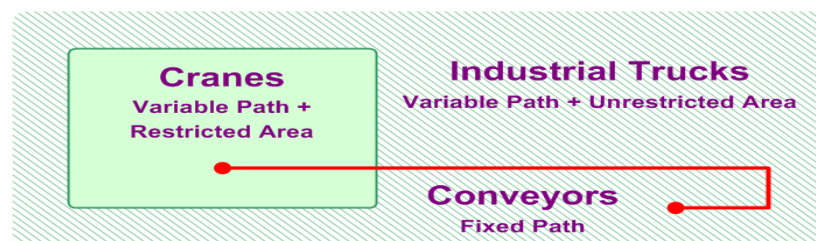


Figure 02 Path and area of operation

OR

- Q- 3 (a)** Maintenance engineering is the discipline and profession of applying engineering concepts for the optimization of equipment, procedures, and departmental budgets to achieve better maintainability, reliability, and availability of equipment? Elaborates. [07]
- Q - 3 (b)** Compile importance of safety engineering. What type of hazards exists in boiler pertaining to risk of explosion, high pressure steam, combustion gases, chemicals, moving machinery and hot surfaces. [07]

SECTION – II

Q - 4 Tata wants employees idle 25% of the time Sample should be accurate within 3% [07]
wants to have 95.45% confidence in the results. Determining the sample size by using the formula $n = \frac{z^2 p(1-p)}{h^2}$

where n = Required sample size
 z = Standard normal deviate for desired confidence level
 p = Estimated value of sample proportion
 h = Acceptable error level in percent
 z = 2 for a 95.45% confidence level

Q - 5 (a) Apply Fishbone (Ishikawa) Diagram (Cause & Effect) to any situation in mechanical [07]
engineering field problem by using different M (Men .Material, Machines, Method
etc.) as shown in figure 03 .One such example is shown for your reference in figure
04.

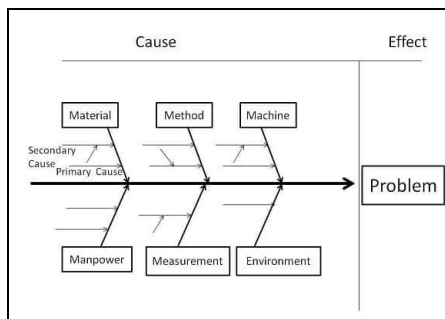


Figure 03 Fishbone Diagram format

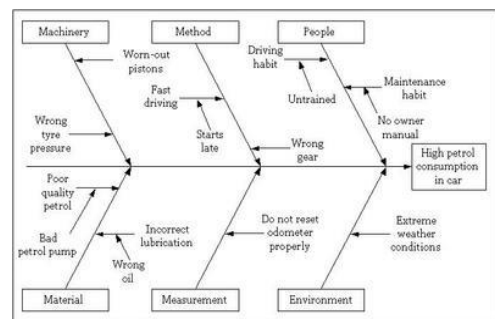


Figure 04 Fishbone Diagram for reference

Q – 5 (b) What do you mean by inventory, types of inventory, inventory control? Enlist steps [07]
involved in inventory control.

OR

Q - 5 (a) Interpret meaning of inventory control model as in figure 05. [07]

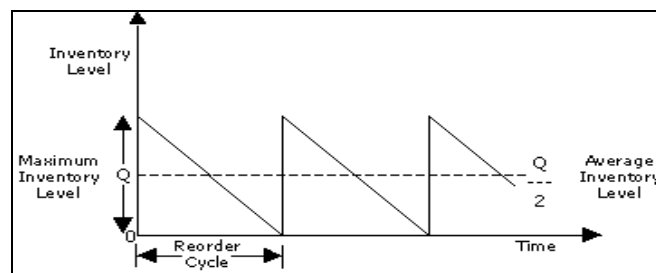


Figure 05 Inventory control model

Q - 5 (b) Point out difference between forging and casting pertaining to cost estimation. [07]

Q - 6 (a) Estimate the time required for making an open tank size $50 \times 50 \times 50$ cm by gas [07]
(Figure 06) welding size of sheets used is $50 \times 40 \times 0.3$ cm. Welding to be done on
inner surface only. Assume fatigue allowances to be 5%. Assume welding speed is 12
min /meter

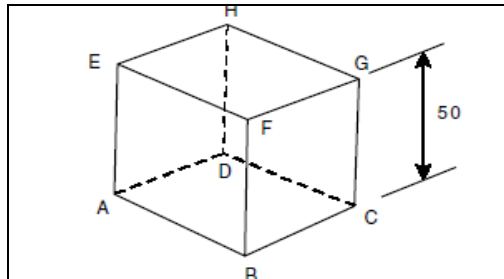


Figure 06 Tank size $50 \times 50 \times 50$ cm

Q-6 (b) Apply why why analysis in given format (Figure 07) to find root cause for any [07]
situation you have undergone in your life.

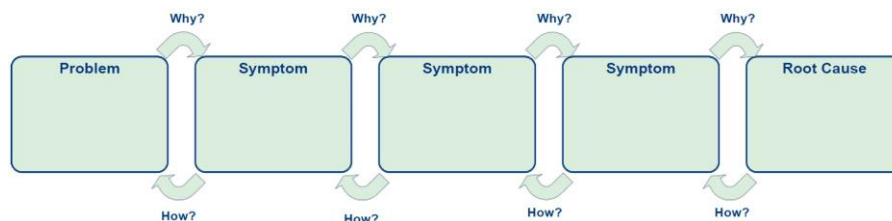


Figure 07 why why analysis format

OR

Q - 6 (a) What are different considerations during estimation of gas welding processes? [07]

(b) Enlist different examination of six sigma certification? What are different [07]
tools/techniques which one should know to get the certificate?
